12.5.4 Application of Integration

Example 12.26

Let the marginal cost function of a firm be $100-10x+0.1x^2$ where x is the output. Obtain the total cost function of the firm under the assumption that its fixed cost is ₹500.

Solution

$$MC = 100 - 10x + 0.1\,x^2$$

$$TC = \int \left(100 - 10x + 0.1x^2\right) dx$$

$$= 100x - 10\frac{x^2}{2} + 0.1\frac{x^3}{3} + c$$

$$= 100x - 5x^2 + \frac{x^3}{30} + c$$

Fixed cost is given as ₹500

$$\therefore TC = 100x - 5x^2 + \frac{x^3}{30} + 500$$

$$= \frac{x^3}{30} - 5x^2 + 100x + 500$$

Example 12.27

The marginal cost function for producing x units is $y = 23 + 16x - 3x^2$ and the total cost for producing zero unit is ₹40. Obtain the total cost function and the average cost function.

Solution:

Given the marginal cost function $y = 23 + 16x - 3x^2$; $c = 40$

₹40 is the fixed cost.

We know that

Total Cost function = $\int$ (Marginal cost function) dx+c

$$TC = \int y\,dx + c$$

$$= \int (23 + 16x - 3x^2)dx + c,$$

where c is a constant

$$= \int 23dx + \int 16x dx - \int 3x^2\,dx + c$$

$$= 23x + 16\left[\frac{x^2}{2}\right] - 3\left[\frac{x^3}{3}\right] + c$$

$TC = 23x + 8x^2 - x^3 + c$

$c = 40$ given

$\therefore TC = 23x + 8x^2 - x^3 + 40$

$$\text{Average cost function} = \frac{TC}{x}$$

$$= 23 + 8x - x^2 + \frac{40}{x}$$

12.5.5 Consumer's Surplus

This theory was developed by the Alfred Marshall. The demand function P(x) reveals the relationship between the quantities that the people would buy at given price. It can be expressed as

$$P = f(x)$$

Consumer surplus is the difference between the price one is willing to pay and the price that is actually paid.

It is represented in the following diagram.

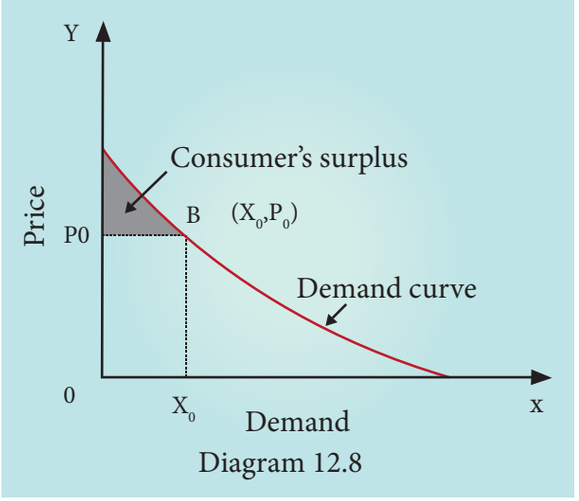


Diagram 12.8

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